

AKASH BAG

Software Engineer | Full-Stack Developer | Agentic AI Practitioner

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Professional Summary

Software Engineer with 2+ years of experience building industrial IoT platforms, analytics dashboards, and scalable full-stack applications using React.js, Next.js, Fastify.js, PostgreSQL, and Docker.

Professional Experience

Software Engineer

Inwork Global Pvt Ltd | Nov 2025 – June 2026

- Architected a distributed query orchestration system handling **500K+** record datasets for analytical filtering, data processing, and CSV export workflows.
- Built async worker-based processing pipelines for long-running queries, achieving sub-second frontend response times and eliminating UI blocking across all report flows.
- Designed dynamic SQL generation systems with JOIN optimization and deduplication strategies, reducing execution time by **~40%** on complex analytical workloads.
- Delivered **2+** analytics and operational dashboards for KPI monitoring, business intelligence, and real-time data visualization.
- Implemented RBAC supporting 4+ user roles with permission-based access control, protected routes, and dynamic UI rendering.
- Leveraged Agentic AI workflows and AI-assisted development to accelerate software delivery by **30%** while maintaining code quality and scalability.

Software Development Engineer L-1

IEM-UEM Group, Kolkata | Sep 2024 – Oct 2025

- Led a team of **4** engineers to architect and deliver a university hiring portal serving **1,000+** students and recruiters, owning architecture, task delegation, and technical decision-making.
- Engineered a real-time industrial IoT monitoring platform for machine health tracking, anomaly detection, and predictive maintenance across **50+** connected industrial devices.
- Built scalable data ingestion pipelines for high-frequency IoT sensor streams with sub-second dashboard refresh rates and 4+ role-based monitoring dashboards.
- Implemented secure API-driven workflows for device management, telemetry processing, and access control across operations and engineering teams.

Software Developer Intern

IEMA Research & Development Pvt Ltd | Mar 2024 – Aug 2024

- Optimized a coding competition platform for a university with **12,000+** students via REST API integrations and real-time leaderboard functionality, improving data refresh latency for live event scoring.
- Developed responsive React.js frontend features contributing to production deployments used by 500+ active users.

Front-End Developer Intern

ReadyToGo Travels Pvt Ltd | May 2022 – Jul 2022

- Developed responsive frontend interfaces and participated in code review processes to maintain UI quality standards.

Technical Skills

Languages: JavaScript, TypeScript, C/C++

Frontend: React.js, Next.js, Tailwind CSS, React Query

Backend: Node.js, Fastify.js, Express.js, Payload CMS

Databases: PostgreSQL, MySQL, MongoDB, Firebase

Tools & DevOps: Drizzle ORM, NextAuth, Docker, Git, GitHub

Architecture: Microservices, RBAC, Distributed Systems

AI & Practices: Agentic AI, AI-Assisted Development, Agile, REST APIs, CI/CD

Education

B.Tech in Computer Science and Technology

Graduated: 2024

University of Engineering & Management, Kolkata

CGPA: 8.8 / 10

Achievements

- Architected a distributed query orchestration system handling **500K+** record datasets, reducing delivery time by **30%** through Agentic AI workflows while maintaining production-grade code quality.
- Contributed to open-source projects across multiple Hacktoberfest events, earning official recognition for community contributions.
- Achieved a max competitive programming rating of **1,704** and solved **200+** problems on LeetCode across multiple coding competitions.

Certifications

Complete React Developer (Redux, Hooks, GraphQL) – Udemy

2023

The Web Developer Bootcamp – Udemy

2023